

Characterizing the γ -ray sky using differentiable probabilistic programming

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Abstract

Analysis of γ -ray data towards the Galactic Center involves a subtle interplay between observations and theoretical models; the complexity of cosmic-ray transport processes in the region, combined with the fact that we only see a two-dimensional projection of this inherently three-dimensional process, means that we typically cannot make precise predictions about crucial emission mechanisms. Consequently, scientific conclusions can depend strongly on specific assumptions made about the modeling of the signal and background, as well as choices involved in the data analysis pipeline. In this work, we use differentiable probabilistic programming, enabled by recent advances in machine learning, in order to account for the large model-space inherent to γ -ray analyses in a principled manner. Focusing on the ongoing mystery of the Galactic Center Excess (GCE), we construct a differentiable forward model and likelihood that liberally uses GPU acceleration and vectorization in order to simultaneously account for a range of possible spatial morphologies consistent with the Excess emission. Our setup allows for efficient inference over the large model space using variational methods. The goal of this work is enable analyses that are more robust to specific modeling choices compared to traditional techniques, and to provide a framework for future analyses.

1. Introduction

Data from the *Fermi* Large Area Telescope (LAT) have revolutionized our understanding of γ -ray astrophysics and continue to provide a wealth of interesting structure to explore. Studying the Inner Galaxy – the central regions of the Milky Way – in γ rays has proved particularly challenging

due to the rich source populations, as well as the frequent interactions of cosmic rays with the high density of gas and radiation in the region. The γ -ray Galactic Center Excess (GCE), first identified over a decade ago (Goodenough & Hooper, 2009) using LAT public data (Atwood et al., 2009), is an excess of photons in the Inner Galaxy whose physical origin is currently unknown. Although the GCE has properties broadly compatible with those expected from annihilating dark matter (DM) (Goodenough & Hooper, 2009; Hooper & Goodenough, 2011; Daylan et al., 2016; Calore et al., 2015; Ajello et al., 2016), competing explanations in terms of a population of unresolved astrophysical point sources (PSs), in particular millisecond pulsars (MSPs), remain viable (Abazajian et al., 2014; Abazajian, 2011). Significant effort has gone into characterizing the GCE and understanding its origin, including through inferring the properties of unresolved PSs in the Inner Galaxy region. A particularly successful method in this direction, Non-Poissonian Template Fitting (NPTF), aims to characterize different populations of unresolved PSs using their photon statistics, via a likelihood-based analysis (Lee et al., 2015; 2016). In the context of the GCE, initial applications of this method showed overwhelming support for the hypothesis of the GCE being sourced by an unresolved population of PSs spatially correlated with the observed GCE morphology (Lee et al., 2016). However, more recent studies showed the method to be susceptible to systematic biases (Leane & Slatyer, 2019; 2020a,b; Chang et al., 2020) partly stemming from degeneracies between different emission components in the region (e.g., those spatially correlated with the GCE, centrally-concentrated stellar populations, and the Galactic disk), with results depending strongly on specific assumptions made about the spatial morphologies of components. *All NPTF analyses so far have assumed inflexible, spatially rigid templates, preventing a rigorous expression and exploration of this degeneracy.*

In this work, we use differentiable probabilistic programming to alleviate this major shortcoming of traditional likelihood-based fitting techniques for analyzing γ -ray data. Using GPU acceleration and vectorization, we construct a forward model for γ -ray emission in the Galactic Center region that can generate high-resolution spatial templates for various emission components, including those associated with the Galactic Center excess signal, on the fly.

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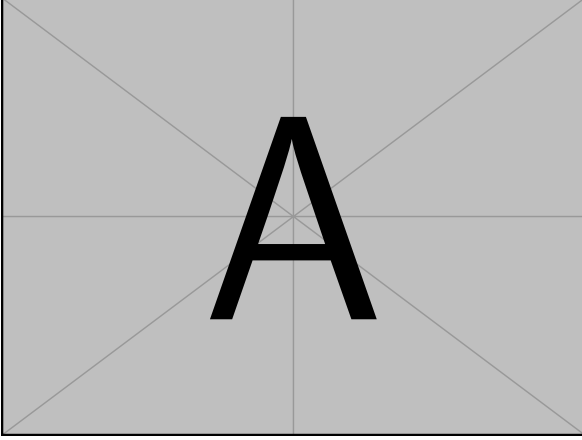


Figure 1. A figure.

This allows for analyses that are robust to specific analysis choices, and opens up the possibility of performing model-independent analyses using the same framework. In the broader context of astrophysics, this work aims to motivate the use of differentiable probabilistic programming as a way to enable high-dimensional Bayesian inference for complex descriptions of diverse astrophysical data.

2. Methodology

We describe our forward model and likelihood, its implementation as a differentiable probabilistic program, as well as the inference strategies we use to do Bayesian posterior inference over the model parameters characterizing various emission mechanisms for Galactic Center γ -ray data.

2.1. The likelihood and forward model

SM: Brief description of regular template fitting as well as NPTF.

SM: Brief description of various templates, and different degrees of freedom.

SM: Specifics of our forward model; hybrid template descriptions.

We implement the NPTF likelihood in the differentiable programming framework Jax (Bradbury et al., 2018). This allows us to include parametric variations on different spatial templates that have been previously included in NPTF analyses, including templates for PSs correlated with the Galactic disk and the GCE, directly as part of the inference pipeline. For example, such variations could include the scale height of the Galactic disk and the slope of the GCE radial profile as additional parameters. Jax implements just-in-time (JIT) compilation and efficient vectorization on GPUs, allowing for fast and real-time generation of high-resolution spatial

templates at inference time. Furthermore, Jax implements automatic batching, allowing for a large number of spatial templates corresponding to different input parameters to be produced at essentially the same computational cost as a single template.

2.2. Inference techniques

Since the entire pipeline is end-to-end differentiable, we can evaluate gradients of the log-likelihood function with respect to the parameters of interest θ characterizing the smooth as well as PS components, $\nabla_{\theta} \log \mathcal{L}(\theta)$. This opens up the possibility of using highly efficient gradient-based posterior inference techniques like Stochastic Variational Inference (SVI) (Hoffman et al., 2012) and Hamiltonian Monte Carlo (HMC) (Hoffman & Gelman, 2011). Unlike traditional sampling techniques employed in the context of *Fermi* GCE analyses so far, these methods can easily scale up to high-dimensional parameter spaces. In particular, variational methods turn posterior inference into an optimization problem by fitting for a flexible functional ansatz on the posterior density and can easily deal with hundreds of parameters. This is in contrast to current public implementations of NPTF, which cannot efficiently include real-time generated templates at inference and are restricted to a modest $\lesssim 30$ parameters with traditional Monte Carlo sampling techniques. We will wrap the constructed likelihood using the probabilistic programming framework NumPyro (Phan et al., 2019), which allows for flexible specification of complex models as well as performing inference on them using gradient-based inference techniques. Illustrative results for the posterior on the GCE and disk template parameters using a preliminary version of the NPTF likelihood written in Jax are shown in The scalability of our implementation will also support inclusion of a wider range of templates for Inner Galaxy PS populations and Galactic diffuse emission, although the development of improved diffuse models is beyond the scope of this proposal.

SM: Brief description of SVI.

3. Experiments

SM: Results of the experiment we will show (posteriors and such).

4. Discussion and Future Work

SM: A summary of our contributions.

- We can easily include multiple signal and background template descriptions in a principled manner.
- Scalability to higher number of parameters.
- Departure from rigid parameters – generate templates



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Figure 2. A figure.

on-the-fly during inference.

- Much faster analysis.

Modern probabilistic programming languages like NumPyro treat distributions and random variables as first-class objects, allowing for the specification of models in a manner that is syntactically close to the PGM. The model can then be directly used for performing inference (by conditioning it on observed data) as well as producing simulations (via sampling the model in the forward direction). The latter will be used to effectively validate the analysis results via posterior-predictive checks, which assess the compatibility between simulated realizations from the data-conditioned model and real data. Strategies such as *enumeration* (Obermeyer et al., 2019), which parallelize the likelihood computation over all possible combinations of the number of individual PSs, will be used to perform explicit marginalization while retaining the benefits of gradient-based posterior inference. This method will have several advantages over all existing approaches. Unlike the NPTF method, it will be able to fully account for pixel-to-pixel correlations. In contrast to machine learning-based approaches, being likelihood-based it will not require the use of black-box feature extractors, and additionally will easily scale to the larger model-spaces described in previous stages of the project since it will not rely on the use of pre-specified training data that is required to have support over the entire model-space.

Directly building on the work done so far, our next goal will

be to characterize the γ -ray PS populations in the Galactic Center in a fully signal model-independent manner, by scanning over a space of spatial *functions* on the sky map, as opposed to using parametric descriptions of signal templates. We will use Gaussian processes to place a prior over this function space, which will allow us to specify a scale on which we expect the morphology of the signal to vary, leaving the per-pixel normalizations otherwise free. The posteriors over both the smooth as well as unresolved PS population templates will provide an important cross check with the fixed-template descriptions used so far, and will, for the first time, deliver a model-independent characterization of the spatial population of unresolved PSs in the Inner Galaxy. A similar Gaussian process-based technique was successfully implemented in Mishra-Sharma & Cranmer (2020) to augment the morphology of the diffuse foreground template. The inclusion of flexible spatial templates in the pipeline is enabled by the highly efficient and differentiable version of the NPTF likelihood.

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